

PSEUDOCODE WRITING

← Back

QUESTIONS

Q1

Student Management System
[BL3 – Apply]
10 marks

Q2

Library Book Management [BL3 – Apply]
10 marks

Q3

Employee Salary Calculation
[BL3 – Apply]
10 marks

Q4

Shape Area Calculator using Polymorphism [BL3 – Apply]
10 marks

Q5

5. Vehicle Registration System
[BL3 – Apply]
10 marks

Q9 9. University Staff Hierarchy [BL4 – Analyze]

10 marks

Analyze the design of a Staff base class extended by TeachingStaff and NonTeachingStaff. Further extend TeachingStaff into Professor and Lecturer (multilevel inheritance). For each level:

- Identify which attributes should be private vs protected
- Override calculateAllowance() at each level
- Analyze how protected access enables inheritance without full exposure Justify the use of multilevel inheritance and explain how access modifiers enforce data hiding.

OOP Concepts: Multilevel Inheritance | Access Modifiers | Data Hiding

Your Answer

CLASS Staff

```
PRIVATE staffId : INTEGER
PRIVATE name : STRING
PRIVATE department : STRING
PROTECTED basicSalary : REAL
```

CONSTRUCTOR Staff(id, n, dept, salary)

```
staffId = id
name = n
```

[BL3 – Apply]
10 marks

Q4

Shape Area Calculator using
Polymorphism [BL3 – Apply]
10 marks

Q5

5. Vehicle Registration System
[BL3 – Apply]
10 marks

Q6

6. Bank Account Hierarchy [BL4
– Analyze]
10 marks

Q7

7. Hospital Patient Record
System [BL4 – Analyze]
10 marks

Q8

8. E-Commerce Product Catalog
[BL4 – Analyze]
10 marks

Q9

9. University Staff Hierarchy
[BL4 – Analyze]
10 marks

Your Answer

```
name = n
department = dept
basicSalary = salary
```

```
ABSTRACT METHOD calculateAllowance() : REAL
END CLASS
```

```
CLASS TeachingStaff EXTENDS Staff
PROTECTED teachingHours : INTEGER
PRIVATE qualification : STRING

CONSTRUCTOR TeachingStaff(id, n, dept, salary, hours, qual)
CALL Staff(id, n, dept, salary)
teachingHours = hours
qualification = qual
```

```
METHOD calculateAllowance() : REAL
RETURN basicSalary * 0.10 + teachingHours * 50
END CLASS
```

```
CLASS Professor EXTENDS TeachingStaff
PRIVATE researchProjects : INTEGER
```

```
CONSTRUCTOR Professor(id, n, dept, salary, hours, qual, projects)
CALL TeachingStaff(id, n, dept, salary, hours, qual)
```

[BL3 – Apply]
10 marks

Q4

Shape Area Calculator using
Polymorphism [BL3 – Apply]
10 marks

Q5

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[BL3 – Apply]
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– Analyze]
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System [BL4 – Analyze]
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[BL4 – Analyze]
10 marks

Q9

9. University Staff Hierarchy
[BL4 – Analyze]
10 marks

Your Answer

```
CALL TeachingStaff(id, n, dept, salary, hours, qual)
researchProjects = projects
```

```
METHOD calculateAllowance() : REAL
    RETURN basicSalary * 0.15 + teachingHours * 60 + researchProjects * 200
END CLASS
```

```
CLASS Lecturer EXTENDS TeachingStaff
    PRIVATE hasLab : BOOLEAN
```

```
CONSTRUCTOR Lecturer(id, n, dept, salary, hours, qual, labFlag)
    CALL TeachingStaff(id, n, dept, salary, hours, qual)
    hasLab = labFlag
```

```
METHOD calculateAllowance() : REAL
    labAmount : REAL
    IF hasLab = TRUE THEN
        labAmount = 5000
    ELSE
        labAmount = 2000
    END IF
    RETURN basicSalary * 0.12 + teachingHours * 40 + labAmount
END CLASS
```

```
CLASS NonTeachingStaff EXTENDS Staff
```


Polymorphism [BL3 – Apply]
10 marks

Q5

5. Vehicle Registration System
[BL3 – Apply]
10 marks

Q6

6. Bank Account Hierarchy [BL4 – Analyze]
10 marks

Q7

7. Hospital Patient Record System [BL4 – Analyze]
10 marks

Q8

8. E-Commerce Product Catalog [BL4 – Analyze]
10 marks

Q9

9. University Staff Hierarchy [BL4 – Analyze]
10 marks

Q10

10. Smart Home Device Controller [BL4 – Analyze]
10 marks

CLASS NonTeachingStaff EXTENDS Staff
PRIVATE adminHours : INTEGER

CONSTRUCTOR NonTeachingStaff(id, n, dept, salary, hours)
CALL Staff(id, n, dept, salary)
adminHours = hours

METHOD calculateAllowance() : REAL
RETURN basicSalary * 0.08 + adminHours * 30
END CLASS

PROCEDURE main()
staffList : ARRAY OF Staff

staffList[0] = NEW Professor(1, "A", "CSE", 50000, 20, "MTech", 3)
staffList[1] = NEW Lecturer(2, "B", "CSE", 40000, 25, "MCA", TRUE)
staffList[2] = NEW NonTeachingStaff(3, "C", "Admin", 35000, 15)

FOR i FROM 0 TO 2
allowance = staffList[i].calculateAllowance()
PRINT allowance
END FOR
END PROCEDURE

 Save Answer